

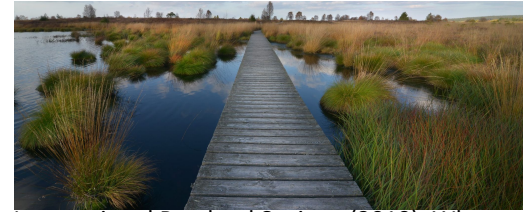
Global Peatland Fire Prediction

A Machine Learning model for Predicting Fire Probability in
Global Peatlands

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Introduction

- Peatlands store up to double the amount of carbon found in every forest on Earth combined, despite them only taking up ~3% of the Earth's surface
- Peatlands have unique fire conditions because their primary fuel source is located in the ground, (eg. variables like soil moisture have higher importance)
- Global fire predictions systems today treat peat-fires like any other wildfire, when important differences in fire conditions exist



International Peatland Society. (2019). What are peatlands? - International Peatland Society. International Peatland Society. <https://peatlands.org/peatlands/what-are-peatlands>



Maps | Global Peatlands Initiative. (2026). Globalpeatlands.org. <https://globalpeatlands.org/story-maps>



Greenpeace International. (2022, October 24). Peatland protection is climate protection. Greenpeace International. <https://www.greenpeace.org/international/story/56176/peatland-protection-is-climate-protection>

Research Questions

- Peatland fires are known to be strongly linked to peat hydrology & soil dryness.

Dry Peat Soil
(highly susceptible
to catching on fire)



Wet Peat Soil
(naturally resistant
to catching on fire)

(Photo Captured by Finalist, 2026)

RQ1. Can we develop a novel global, daily peatland-specific fire prediction system that accounts for important peat hydrology & dryness using machine learning?

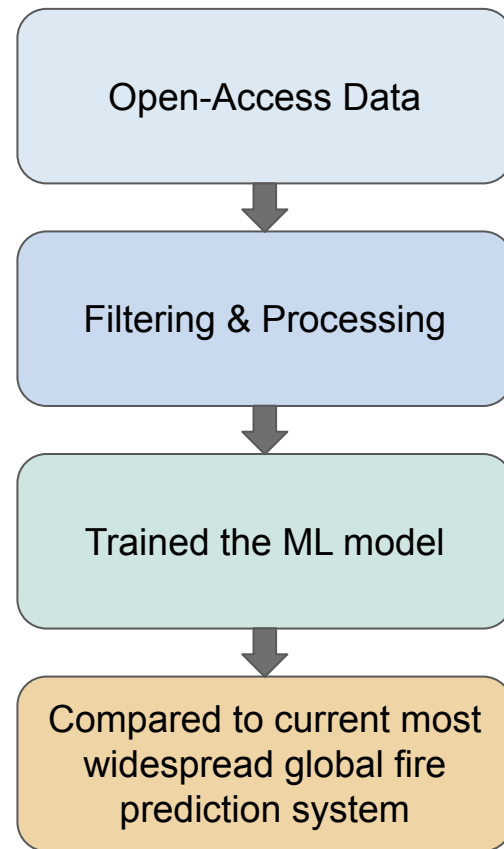
RQ2. Can this system outperform the Canadian Fire Weather Index (FWI), one of the most widely used operational fire-danger indices worldwide?

Methods - What did you do?

I developed a novel machine learning model to specifically predict daily chances of fire in peatlands worldwide.

- **Data Sourcing:** Open-access satellite reanalysis, population, and road datasets are used
- **Data Processing:** Removed low-confidence fire detections and filtered out static hotspots
- **Training & Prediction:** Tested multiple approaches and selected a fine-tuned XGBoost machine learning model

Benchmarking: This system was compared against the widespread Canadian Fire Weather Index (FWI), which is used today for fire prediction in dozens of countries



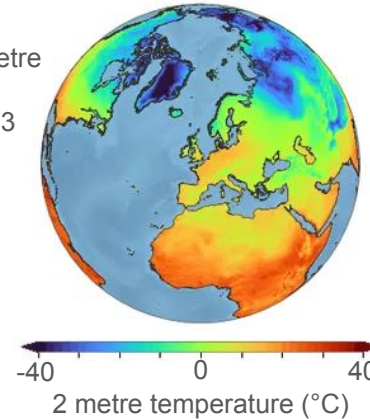
(Graphic created by Finalist using powerpoint, 2026)

Data Sourcing

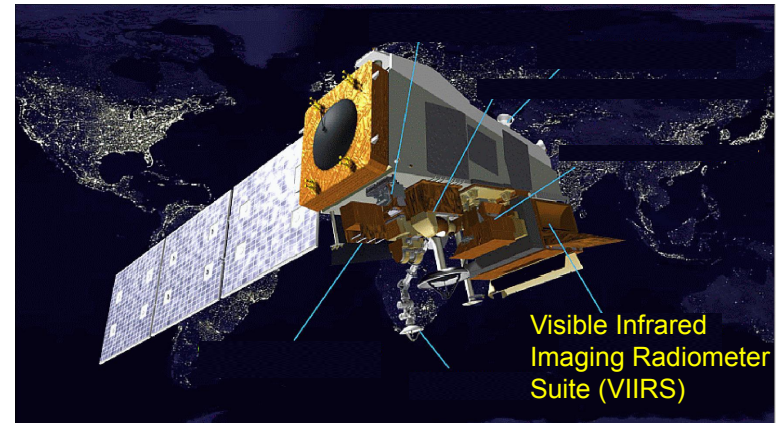
- Peatland hydrology & dryness is accounted for, as well as unique drought memory variables
- **26 independent variables** for today and yesterday are fed to the AI system to capture many details
- The AI system was trained to predict fire detections captured by **infrared & visible light sensors** aboard NASA & NOAA satellites

Muñoz Sabater, J. (2019): ERA5-Land hourly data from 1950 to present. Copernicus Climate Change Service (C3S) Climate Data Store (CDS). DOI: 10.24381/cds.e2161bac (Accessed on 04-04-2026)

ERA5-Land 2 metre
temperature
1 January 2023



- Example taken from ERA5-Land satellite reanalysis dataset



JPSS-1 (Joint Polar Satellite System-1) / NOAA-20 and JPSS-2 series. (2025). Eoportal.org. https://www.eoportal.org/ftp/satellite-missions/j/JPSS1_200122/JPSS1.html

Data Processing

1. Static hotspots, missing days, and corrupt values were removed
2. The geospatial data was reformatted to the same resolution
3. Non-peatlands were masked out by NASA's SMAP peatland mask
4. Data was optimized for storage and loading into RAM/VRAM

All shaded pixels are peatlands

***Example in Europe/Russia**

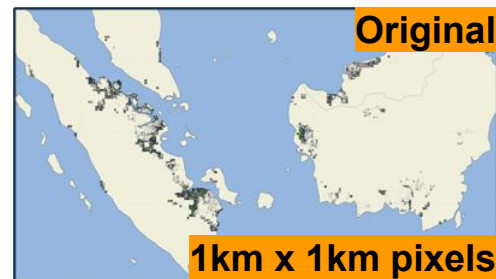
Red pixels were removed for not being vegetation fires



(Image created by Finalist using Python & Cartopy, 2026)

Population data in SE Asia
is regridded to needed
resolution

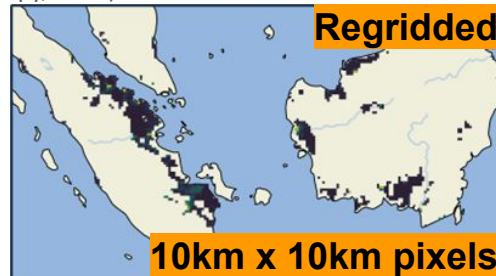
***Example in SE Asia**



(Image created by
Finalist using Python &
Cartopy, 2026)

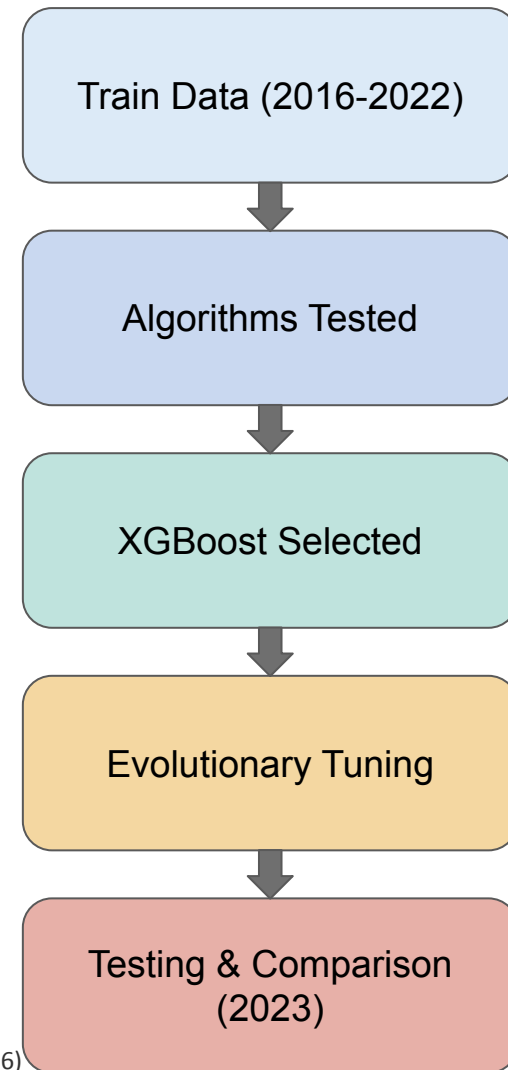


(Image created by
Finalist using Python &
Cartopy, 2026)



Training & Prediction

- Multiple Machine Learning Algorithms were tested to identify best candidate
 - The eXtreme Gradient Boosting (XGBoost) algorithm was selected due to high performance and computational efficiency
- The XGBoost model was trained to predict peatlands fires on the years (2016-2022)
- The model's hyperparameters were fine-tuned through an evolutionary search algorithm
- The model was tested & evaluated against the Widespread Canadian Fire Weather Index on the year (2023)



Results Visualized

- The AI model produced sharper, more localized fire-risk predictions than the widely used Canadian Fire Weather Index (FWI).
- The AI model was also found to perform better than the FWI across most peatland-rich regions

Images shown are only a regionally cropped case study; global predictions are able to be visualized

Case study shown

Tropical Americas, May 1, 2023

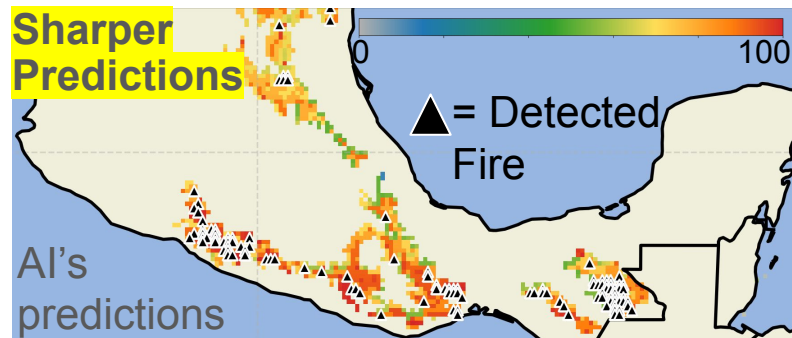
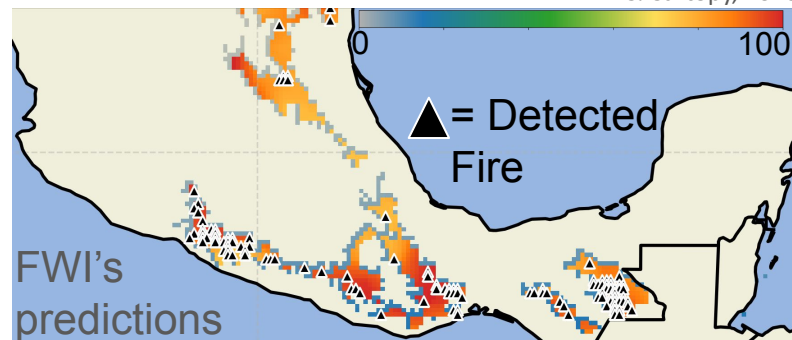
The top fire-detection day across all regions during 2023 testing

*Colored pixels = Peatlands

*Warmer colors = higher predictions

*Cooler colors = lower predictions

(Image created by
Finalist using Python
& Cartopy, 2026)



(Image created by
Finalist using Python
& Cartopy, 2026)

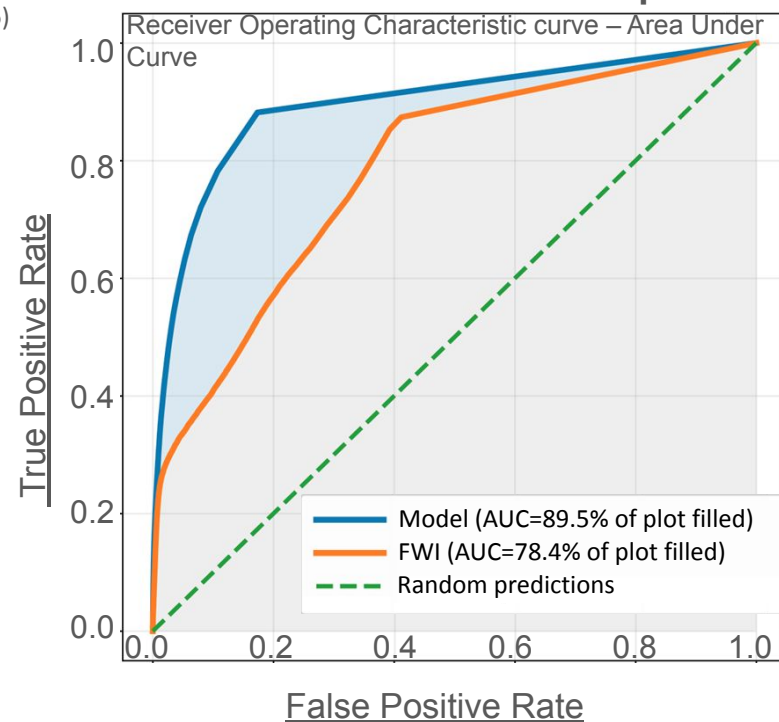
Testing Metrics

(Graph created by
Finalist using Python &
Matplotlib, 2026)

- The AI model scored 11.1% higher than the Canadian Fire Weather Index (FWI) in distinguishing peatland fire-prone conditions globally.
 - *Global ROC-AUC: AI = 89.5%, FWI = 78.4%*
- The model's predicted chances of fire are highly reliable and are accurate to within 0.04% on average compared to observed fire detections.
 - *Global Expected Calibration Error (ECE): 0.000493*

ROC-AUC: how well the model ranks fire above non-fire cases
ECE: how closely predicted probabilities match real outcomes

Global ROC-AUC Curve Comparison



- The AI model achieved a better tradeoff between true positives and false positives than the FWI across decision thresholds.

Findings

Summary of Results:

- A machine learning model that predicts fires specifically for peatlands and utilizes factors that are important to them such as hydrology & dryness performs highly on the global scale.

Evaluation:

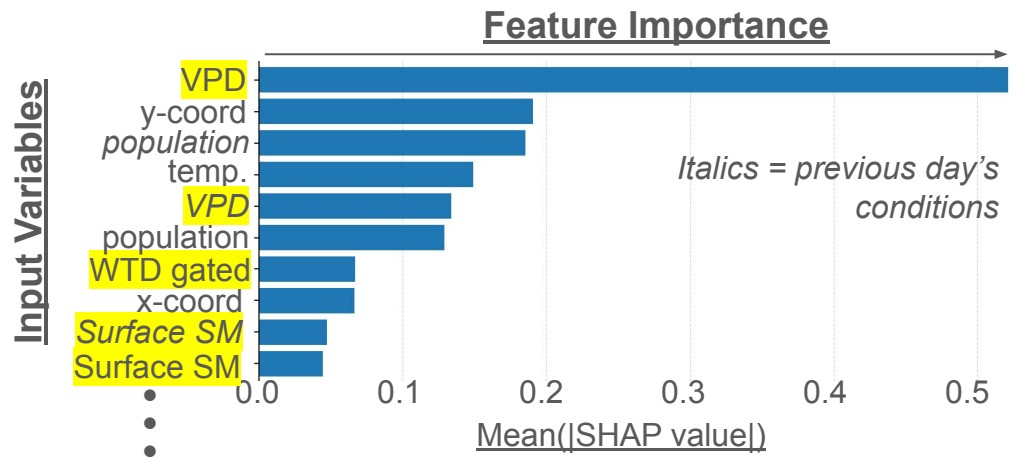
- The AI system was evaluated globally and across many defined peatland regions in 2023.
- It predicted chances of fire to within ~0.04% on average demonstrating high reliability.
- It separated fire from non-fire conditions 89.5% of the time globally. Which is 11.1% higher than the Canadian Fire Weather Index, one of the most widely used fire indices worldwide.

(Graph created by
Finalist using
Python &
Matplotlib, 2026)

Statistical Analysis:

- The model prioritized factors theorized to be important. Eg. dryness (VPD), water-table depth (WTD gated), and soil moisture (surface SM).

(Graphic created by Finalist
using powerpoint, 2026)



Conclusion

Answering Research Questions

- The findings show that a global, daily, peatland-specific XGBoost model that accounts for peat hydrology and dryness can improve peatland fire prediction over the Canadian Fire Weather Index globally whilst performing reliably.

Further Challenges

- To develop my model into a proper forecasting system, I will need a researcher's license from the European Centre for Medium-Range Weather Forecasts to gain access to weather forecasts from their High-Resolution Integrated Forecasting System. These forecasts will then be fed to my model to make it forecast fires into the future.

Practical Application

- Can help many more communities worldwide to accurately predict & prevent peatland fires without relying on generic wildfire systems. Reducing carbon emissions & hazardous air pollution, helping to conserve one of Earth's most important carbon sinks.

Key References

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